

QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9
10        System.out.println("Sum = " + (a + b));
11        System.out.println("Difference = " + (a - b));
12        System.out.println("Product = " + (a * b));
13
14        if (b != 0) {
15            System.out.println("Quotient = " + (a / b));
16        } else {
17            System.out.println("Quotient = Cannot divide by zero");
18        }
19
20        sc.close();
21    }
```

OUTPUT

```
Sum = 15
Difference = 5
Product = 50
Quotient = 2
```

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class maxofthree {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter first number: ");
6         int a = sc.nextInt();
7         System.out.print("Enter second number: ");
8         int b = sc.nextInt();
9         System.out.print("Enter third number: ");
10        int c = sc.nextInt();
11        if (a >= b && a >= c) {
12            System.out.println("Maximum number is: " + a);
13        } else if (b >= a && b >= c) {
14            System.out.println("Maximum number is: " + b);
15        } else {
16            System.out.println("Maximum number is: " + c);
17        }
18        sc.close();
19    }
20 }
```

OUTPUT

```
Enter first number: Enter second number: Enter third number: Maximum number
is: 28
```

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class LeapYear {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter a year: ");
6         int year = sc.nextInt();
7         if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
8             System.out.println(year + " is a Leap Year. ");
9         } else {
10            System.out.println(year + "is not a Leap Year.");
11        }
12        sc.close();
13    }
14 }
```

OUTPUT

Enter a year: 2024 is a Leap Year.

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class FibonacciSeries {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter the limit: ");
6         int limit = sc.nextInt();
7         int first = 0, second = 1;
8         System.out.println("Fibonacci series up to" + limit + ": ");
9         while (first <= limit) {
10            System.out.print(first + " ");
11            int next = first + second;
12            first = second;
13            second = next;
14        }
15        sc.close();
16    }
17 }
```

OUTPUT

Enter the limit: Fibonacci series up to20:
0 1 1 2 3 5 8 13

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class factorial {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter a number: ");
6         int num = sc.nextInt();
7         long factorial = 1;
8         for (int i = 1; i <= num; i++) {
9             factorial = factorial * i;
10        }
11        System.out.println("Factorial of " + num + " = " + factorial);
12        sc.close();
13    }
14 }
```

OUTPUT

```
Enter a number: Factorial of 5 = 120
```

QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:*

```
**
***
****
*****
```

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3
4         for (int i = 2; i <= 5; i++) {
5             for (int j = 1; j <= i; j++) {
6                 System.out.print("*");
7             }
8             System.out.println();
9         }
10
11    }
12 }
```

OUTPUT

```
**
***
****
*****
```

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] arr = new int[n];
9
10        for (int i = 0; i < n; i++) {
11            arr[i] = sc.nextInt();
12        }
13
14        int key = sc.nextInt();
15
16        int low = 0, high = n - 1;
17        int found = -1;
18
19        while (low <= high) {
20            int mid = (low + high) / 2;
21
```

OUTPUT

Element found at index 2

QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class GCD {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter first number: ");
6         int num1 = sc.nextInt();
7         System.out.print("Enter second number: ");
8         int num2 = sc.nextInt();
9         int gcd = 1;
10        for (int i = 1; i <= num1 && i <= num2; i++) {
11            if (num1 % i == 0 && num2 % i == 0) {
12                gcd = i;
13            }
14        }
15        System.out.println("GCD of " + num1 + " and " + num2 + " is: " + gcd);
16        sc.close();
17    }
18 }
```

OUTPUT

Enter first number: Enter second number: GCD of 24 and 36 is: 12

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int count = 0;
9
10        if (n <= 1) {
11            System.out.println("Not Prime");
12        } else {
13            for (int i = 1; i <= n; i++) {
14                if (n % i == 0) {
15                    count++;
16                }
17            }
18
19            if (count == 2) {
20                System.out.println("Prime");
21            } else {
```

OUTPUT

Prime

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter number of elements: ");
8         int n = sc.nextInt();
9
10        int[] a = new int[n];
11
12        System.out.println("Enter the elements:");
13        for (int i = 0; i < n; i++) {
14            a[i] = sc.nextInt();
15        }
16
17        // Selection Sort
18        for (int i = 0; i < n - 1; i++) {
19            int min = i;
20
21            for (int j = i + 1; j < n; j++) {
```

OUTPUT

Enter number of elements: Enter the elements:
Sorted array:
11 12 22 25 64

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Table {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         // Input
8         System.out.print("Enter a number: ");
9         int n = sc.nextInt();
10
11        // Multiplication Table
12        System.out.println("Multiplication Table of " + n + ":");
13        for (int i = 1; i <= 10; i++) {
14            System.out.println(n + " x " + i + " = " + (n * i));
15        }
16
17        sc.close();
18    }
19 }
```

OUTPUT

```
Enter a number: Multiplication Table of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
```

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     static int factorial(int n) {
6         if (n == 0 || n == 1)
7             return 1;
8         else
9             return n * factorial(n - 1);
10    }
11
12    public static void main(String[] args) {
13        Scanner sc = new Scanner(System.in);
14
15        int n = sc.nextInt();
16
17        System.out.println(factorial(n));
18
19        sc.close();
20    }
21 }
```

OUTPUT

120

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

```
*****
****
***
**
*
```

PROGRAM EDITOR

Java

Run

Save

```
1 public class LTPattern {
2     public static void main(String[] args) {
3
4         for (int i = 5; i >= 1; i--) {
5             for (int j = 1; j <= i; j++) {
6                 System.out.print("*");
7             }
8             System.out.println();
9         }
10
11     }
12 }
```

OUTPUT

```
*****
****
***
**
*
```

QUESTIONS 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class LCM {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter first number: ");
8         int a = sc.nextInt();
9
10        System.out.print("Enter second number: ");
11        int b = sc.nextInt();
12
13        int max = (a > b) ? a : b;
14
15        while (true) {
16            if (max % a == 0 && max % b == 0) {
17                System.out.println("LCM = " + max);
18                break;
19            }
20            max++;
21        }
22    }
```

OUTPUT

```
Enter first number: Enter second number: LCM = 36
```

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class DecimalToBinary {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter a decimal number: ");
8         int num = sc.nextInt();
9
10        int binary = 0, place = 1;
11
12        while (num > 0) {
13            int rem = num % 2;
14            binary = binary + (rem * place);
15            place = place * 10;
16            num = num / 2;
17        }
18
19        System.out.println("Binary Number = " + binary);
20
21        sc.close();
22    }
23 }
```

OUTPUT

```
Enter a decimal number: Binary Number = 1010
```

QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class BinaryToDecimal {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter a binary number: ");
8         int binary = sc.nextInt();
9
10        int decimal = 0;
11        int base = 1;
12
13        while (binary > 0) {
14            int lastDigit = binary % 10;
15            decimal = decimal + (lastDigit * base);
16            base = base * 2;
17            binary = binary / 10;
18        }
19
20        System.out.println("Decimal Number = " + decimal);
21
22    }
23 }
```

OUTPUT

```
Enter a binary number: Decimal Number = 10
```

QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class SumOfDigits {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter a number: ");
8         int num = sc.nextInt();
9
10        int sum = 0;
11
12        while (num > 0) {
13            int digit = num % 10;
14            sum = sum + digit;
15            num = num / 10;
16        }
17
18        System.out.println("Sum of digits = " + sum);
19
20        sc.close();
21    }
```

OUTPUT

Enter a number: Sum of digits = 10

QUESTIONS 31 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class ProductOfDigits {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter a number: ");
8         int num = sc.nextInt();
9
10        int product = 1;
11
12        while (num > 0) {
13            int digit = num % 10;
14            product = product * digit;
15            num = num / 10;
16        }
17
18        System.out.println("Product of digits = " + product);
19
20        sc.close();
21    }
```

OUTPUT

Enter a number: Product of digits = 24

QUESTIONS 31 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Power {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter base: ");
8         int base = sc.nextInt();
9
10        System.out.print("Enter exponent: ");
11        int exponent = sc.nextInt();
12
13        int result = 1;
14
15        for (int i = 1; i <= exponent; i++) {
16            result = result * base;
17        }
18
19        System.out.println("Power = " + result);
20
21        sc.close();
22    }
23 }
```

OUTPUT

```
Enter base: Enter exponent: Power = 32
```

QUESTIONS 31 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class AreaCircle {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter the radius: ");
8         double radius = sc.nextDouble();
9
10        double area = 3.14159 * radius * radius;
11
12        System.out.println("Area of Circle = " + area);
13
14        sc.close();
15    }
16 }
```

OUTPUT

```
Enter the radius: Area of Circle = 78.53975
```

QUESTIONS 31 / 31 completed

Sphere

SPHERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Sphere {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter the radius: ");
8         double radius = sc.nextDouble();
9
10        double volume = (4.0 / 3.0) * 3.14159 * radius * radius * radius;
11
12        System.out.println("Volume of Sphere = " + volume);
13
14        sc.close();
15    }
16 }
```

OUTPUT

```
Enter the radius: Volume of Sphere = 523.5983333333332
```

QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Palindrome {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter a string: ");
8         String str = sc.nextLine();
9
10        String rev = "";
11
12        for (int i = str.length() - 1; i >= 0; i--) {
13            rev = rev + str.charAt(i);
14        }
15
16        if (str.equals(rev)) {
17            System.out.println("Palindrome");
18        } else {
19            System.out.println("Not a Palindrome");
20        }
21    }
```

OUTPUT

```
Enter a string: Palindrome
```

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Armstrong {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter a number: ");
8         int num = sc.nextInt();
9
10        int original = num;
11        int sum = 0;
12
13        while (num > 0) {
14            int digit = num % 10;
15            sum = sum + (digit * digit * digit);
16            num = num / 10;
17        }
18
19        if (sum == original) {
20            System.out.println("Armstrong Number");
21        } else {
```

OUTPUT

```
Enter a number: Armstrong Number
```

QUESTIONS 31 / 31 completed

RT-num.Pattern

RT- NUM.PATTERN

Write a Java program to print the following pattern using loops:

```
1
22
333
4444
55555
```

PROGRAM EDITOR

Java

Run

Save

```
1 public class RTNumPattern {
2     public static void main(String[] args) {
3
4         for (int i = 1; i <= 5; i++) {
5             for (int j = 1; j <= i; j++) {
6                 System.out.print(i);
7             }
8             System.out.println();
9         }
10    }
11 }
12 }
```

OUTPUT

```
1
22
333
4444
55555
```

QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class PerfectNumber {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter a number: ");
8         int num = sc.nextInt();
9
10        int sum = 0;
11
12        for (int i = 1; i < num; i++) {
13            if (num % i == 0) {
14                sum += i;
15            }
16        }
17
18        if (sum == num) {
19            System.out.println(num + " is a Perfect Number");
20        } else {
21            System.out.println(num + " is not a Perfect Number");
22        }
23    }
24 }
```

OUTPUT

Enter a number: 6 is a Perfect Number

QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter the number of elements: ");
8         int n = sc.nextInt();
9
10        int[] arr = new int[n];
11
12        System.out.println("Enter the array elements:");
13        for (int i = 0; i < n; i++) {
14            arr[i] = sc.nextInt();
15        }
16
17        int max = arr[0];
18
19        for (int i = 1; i < n; i++) {
20            if (arr[i] > max) {
21                max = arr[i];
22            }
23        }
24    }
25 }
```

OUTPUT

Enter the number of elements: Enter the array elements:
Largest element = 89

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++) {
11            a[i] = sc.nextInt();
12        }
13
14        int max = Integer.MIN_VALUE;
15        int secondMax = Integer.MIN_VALUE;
16
17        for (int i = 0; i < n; i++) {
18            if (a[i] > max) {
19                secondMax = max;
20                max = a[i];
21            } else if (a[i] > secondMax && a[i] != max) {
```

OUTPUT

25

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] arr = new int[n];
9
10        for (int i = 0; i < n; i++) {
11            arr[i] = sc.nextInt();
12        }
13
14        int min = arr[0];
15
16        for (int i = 1; i < n; i++) {
17            if (arr[i] < min) {
18                min = arr[i];
19            }
20        }
21    }
```

OUTPUT

3

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++) {
11            a[i] = sc.nextInt();
12        }
13
14        int min = Integer.MAX_VALUE;
15        int secondMin = Integer.MAX_VALUE;
16
17        for (int i = 0; i < n; i++) {
18            if (a[i] < min) {
19                secondMin = min;
20                min = a[i];
21            } else if (a[i] < secondMin && a[i] != min) {
```

OUTPUT

2

QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++) {
11            a[i] = sc.nextInt();
12        }
13
14        // Bubble Sort
15        for (int i = 0; i < n - 1; i++) {
16            for (int j = 0; j < n - 1 - i; j++) {
17                if (a[j] > a[j + 1]) {
18                    int temp = a[j];
19                    a[j] = a[j + 1];
20                    a[j + 1] = temp;
21                }
22            }
23        }
24    }
25 }
```

OUTPUT

12 22 25 34 64

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] arr = new int[n];
9
10        for (int i = 0; i < n; i++) {
11            arr[i] = sc.nextInt();
12        }
13
14        // Insertion Sort
15        for (int i = 1; i < n; i++) {
16            int key = arr[i];
17            int j = i - 1;
18
19            while (j >= 0 && arr[j] > key) {
20                arr[j + 1] = arr[j];
21                j--;
```

OUTPUT

12 22 25 34 64